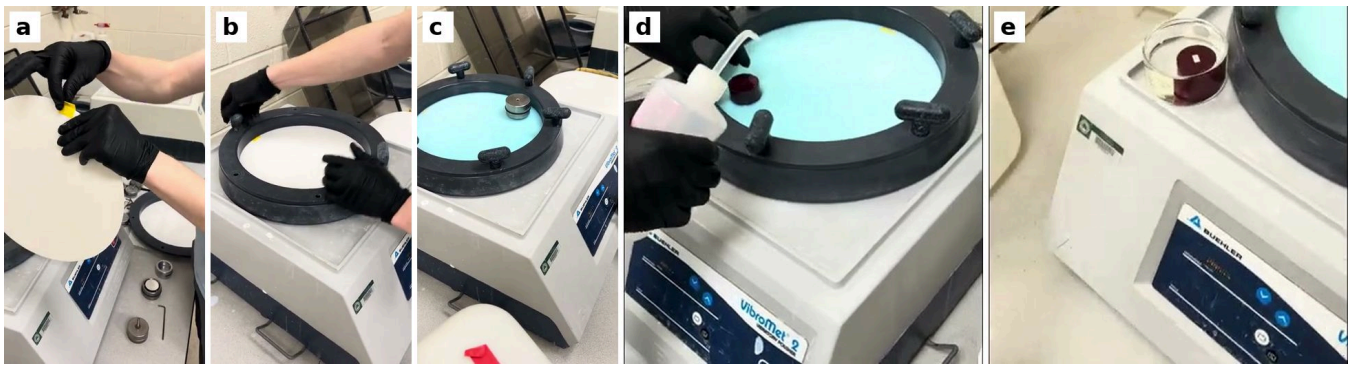
Vibratory Polisher SOP: 0.1 µm Diamond (Buehler VibroMet 2)

Draft, 2026-11-03. Nobody has run the VibroMet with diamond yet, so expect to change this after the first sample.

This is Gage's colloidal silica procedure with the silica swapped for 0.1 µm diamond. His procedure comes from two videos of the May AlSi10Mg sample (setup and removal and cleaning) and his cleaning sheet. The diamond changes come from the VibroMet 2 manual, PACE's guides and two Edison literature searches, listed under Sources.

It replaces Steps 7 and 8 of the polishing SOP, the diamond film and the clean after it. Start with a puck that has been through Step 6 (1 µm alumina, then the methanol ultrasonic). When you're done here, go on to Step 9 (Kapton tape and the saw).

> **No constant spray with diamond.** Colloidal silica had to stay wet because it sets as it dries: as the water leaves, the silica particles bond to each other and to the face, and the layer won't wash off. Drying causes it, not oxygen. Diamond doesn't set. Dried diamond suspension leaves loose diamond and a trace of the water-soluble additives, and both wash off. Edison's search found no study of dried diamond residue on aluminum, so this rests on the chemistry. Its advice: "Prompt but not frantic: aim to begin the water rinse within roughly a minute and never let a thick slurry puddle dry." You can also take the puck out partway through to check it, which you couldn't do with silica.
>
> **The risk to watch for is embedded diamond, not drying.** Buehler polished cast Al-Si alloys with 9 and 1 µm diamond suspensions and found the surfaces "heavily embedded with diamond abrasive", mostly in the aluminum. Diamond paste didn't embed, and Buehler concluded "embedment is mainly a problem with fine diamond particles" (Vander Voort, Buehler Tech-Notes 3(2)). Edison found no published run of a diamond suspension this fine on aluminum, in a vibratory polisher or otherwise. Embedded diamond reads as C in EDS. So keep the load light and the run short, and check the first sample for it at the SEM (Step 8).



Frames from Gage's videos of the colloidal silica run: (a) tape pull-tab on the back of a new cloth, (b) clamp ring going on, (c) the specimen holder with one weight, running, (d) spraying the puck as it comes off the cloth, (e) the puck face up under ethanol.

What you need

Item	Notes
0.1 µm polycrystalline diamond, water-based (PACE DIAMAT `WPC-0110`)	PACE confirmed it has no silica or alumina. Water based, so it rinses off with water
A new 12" adhesive final polishing cloth	Allied Red Final C (`90-150-355`, on hand) until the Mark V pad arrives. Never one that has had colloidal silica or alumina on it
DI water squirt bottle	DI only; tap water dries to Ca and Mg spots
Ethanol squirt bottle	200 proof. This is what Gage sprayed
Two clean glass dishes	Deep enough to cover the puck, as in Gage's sheet
Electroplated tape	For the pull tab on the cloth
Lint-free wipes, paper towels, gloves	
Ultrasonic cleaner, optical microscope	

Step 1

Clean the VibroMet. Other people use it with colloidal silica, and titanium runs add hydrogen peroxide. Any silica left in the bowl ends up on your sample as Si.

- If someone else's cloth is in the bowl, don't touch it. Ask them, or wait.
- Clean the bowl with water and paper towels until all loose material is out. Look hard at the rim, the clamp ring and under the ring, where silica dries into a white crust. Then wipe it all down with ethanol. Gage did water, then ethanol.
- The bowl has to be clean and **dry** before an adhesive cloth goes in (VibroMet 2 manual, p. 9).
- Clean the specimen holder and the weight the same way, water then ethanol, including the bottom face that sits on the cloth.

Step 2

Put in a new cloth.

- Peel the backing off and keep it, so it can go back on for storage. Stick a tab of electroplated tape to the back near the edge, so the cloth can be pulled out later (photo a).
- Lay the cloth in the bowl and press it flat from the center out, so no air is trapped under it.
- Put the clamp ring on and tighten its knobs (photo b).

- Write on the backing and the bag: "0.1 µm diamond, AlSi10Mg only" and the date. One cloth per abrasive and alloy, like the pads in Step 5 of the polishing SOP.

Step 3

Charge the cloth with diamond.

- Look at the bottle first. Gage found dried crust around the colloidal silica cap ("those chunks are bigger than 0.05"). Wipe the cap and the neck with a damp lint-free wipe, pulling away from the opening.
- If there's sediment at the bottom of the bottle, turn the closed bottle over gently a few times to mix it back in. Diamond can settle out; colloidal silica doesn't.
- Wet the cloth thoroughly with DI water first (manual, p. 12).
- Pour the diamond on and spread it with a clean gloved finger until the whole cloth is covered with a film of liquid (manual, p. 12). Gage used 200 mL of silica, "the amount to cover the bottom". Diamond costs more, so start with **about 50 mL** poured over the wet cloth, spread it, and add more diamond until the whole cloth has a film of liquid. Write down how much it took. PACE sells this suspension ready to use, so don't thin it with extra water; only add DI water later to replace what evaporates. If the first check shows little progress, add more diamond; PACE's fix for a slow polish is to "recharge pad with diamond".

Step 4

Load the puck and start.

- Slide the puck into the specimen holder, face down, and tighten the hex screw.
- Wet the bottom of the holder with DI water before you set it on the cloth. The manual says this prevents chatter (p. 11).
- Put the weight on the holder. Gage used one weight and never needed more ("you can actually screw in a post and add more weights, but I've never felt like I needed to"). Keep it at one for aluminum, since more force means more diamond pressed into the metal.
- Put the lid on. Press the power button, then raise the amplitude. Gage starts at 50% and watches: the holder should travel smoothly around the bowl in a circle. If it hits the wall, turn the amplitude down.

Step 5

Check it every 30 minutes, and stop when the alumina scratches are gone.

- The manual says to check every 15-30 minutes and that "some polishing times will exceed one hour" (p. 13). Gage ran silica for 4 hours, and 6 on a slow day. Edison found no published time for this step and suggests a first trial of 1-2 hours, checked every 30 minutes. Check rather than guess.
- To check: stop the machine and lift the holder out. Rinse the face with DI water, then ethanol, and dry it. Look at it under the optical microscope at the highest magnification.

Before putting it back, rinse the bottom of the holder with DI water. Keep the lid on the bowl while you're away.

- Stop when the scratches from the 1 µm alumina are gone. Don't keep going after that. On a napped cloth, extra time polishes the soft aluminum away from the hard Si, which leaves the Si standing proud.
- If scratches are still there after 4 hours, they're deeper than the alumina leaves, probably from the 1200 paper. Go back to Step 5 of the polishing SOP instead of running longer.
- If the cloth starts to look dry on a long run, add DI water. Keep the lid on so it evaporates slowly.

Step 6

Take the puck out and clean it.

- Set up before you stop the machine: the DI water and ethanol squirt bottles, two clean glass dishes deep enough to cover the puck (one with DI water, one with ethanol), and the ultrasonic, cleaned and filled with DI water.
- Stop the machine and lift the holder off. Rinse the face generously with DI water within about a minute. Water is what dissolves the suspension's additives; alcohol only pushes the water off afterwards. There's no need for the non-stop stream the silica needed.
- Take the puck out of the holder and put it face up in the dish of DI water. Cover the dish (lid or parafilm, as Gage did) and stand it in the ultrasonic for 2 minutes. PACE's cleaning guide gives 1-3 minutes.
- Move the puck to the dish of ethanol, cover it, and give it 2 more minutes. Methanol also works, as in Step 6 of the polishing SOP, but Edison found nothing that needs methanol in particular, and ethanol is less toxic. Alcohol always stays in the covered dish, never straight in the tank, because it's flammable.
- Dry it with compressed air, sweeping across the face. Gage: "I always just blow right along the sample, and the sides just have to dry slow." Aim to be dry within a few minutes of lifting the puck out. Never leave it sitting in water.
- Look at it under the optical microscope. If there's haze or specks, repeat both baths. If they're still there, add a drop of neutral lab detergent to the water bath for one more round, then rinse with DI water and finish with the ethanol bath. Detergent isn't routine because it leaves its own residue.
- Then go straight on to Step 9 of the polishing SOP. If you have to stop, put the dry puck in the vacuum jar.

Step 7

Clean up.

- Soak up the free liquid in the bowl with paper towels. PACE's safety data sheet lists no hazardous ingredients but says to dispose of it according to local rules, so ask the lab manager before pouring larger amounts down the sink.

- Rinse the cloth with DI water. If nobody else needs the bowl, leave the cloth in it with the lid on. Otherwise pull it out by the tab, put the backing back on and bag it. Allied says each removal weakens the adhesive, so replace a cloth that no longer lies flat.
- Wash the bowl, the clamp ring, the holder and the weight with water, then wipe them with ethanol, as in Step 1.

Step 8

Check the first diamond sample at the SEM before trusting the method.

- Take BSE images at high magnification. Embedded diamond is carbon, so it shows as dark specks, mostly in the aluminum between the Si.
- Take SE images too. Bright, raised Si and rounded pore edges mean relief from too long a run.
- At 5 kV, compare the C peak with the pure Al standard, polished and cleaned the same way on the same day. Keep C and O in the element list.
- If it's heavily embedded, don't keep polishing longer or harder. Tell the thread, and ask Mike what he does (see Open questions).

Colloidal silica and diamond, side by side

	Gage's colloidal silica run (May)	This SOP (0.1 µm diamond)
Abrasive	0.05 µm colloidal silica, 200 mL	0.1 µm water-based polycrystalline diamond, `WPC-0110`
Why it polishes	Chemical and mechanical: the alkaline silica also attacks the metal	Mechanical only
Time	4 h, 6 h that day	Until the alumina scratches are gone, checked every 30 min
Checking partway	Not practical: every check means the full clean	Rinse, dry, look, put it back
Taking it out	Never let air reach the face: a non-stop ethanol stream, straight into a bowl of ethanol	Rinse with DI water, then ethanol, within about a minute
Cleaning	Acetone 10 min, ethanol 8 min, methanol 5 min, all ultrasonic	DI water rinse, then 2 min ultrasonic in DI water and 2 min in ethanol
What can go wrong	Dried silica, which reads as Si and O in EDS	Embedded diamond, which reads as C. Buehler found fine diamond suspensions embed in Al-Si

Open questions

- Ask Mike what he runs on the vibratory polisher for aluminum with his 0.1 µm diamond: time, weight, cloth, and whether he has seen embedded diamond. He suggested the slurry,

and Edison found no published procedure to compare against.

- How much suspension does a charge take? Write down what the first run used. If the bottle is the 250 mL size, that may only be a few charges.
- Which Mark V pad was ordered (black `CPS-12A` or `VC12`)? See the 11-02 comparison.
- PACE's safety data sheet gives the pH as both "7~9" and "9-11" (Section 9). Check it with a pH strip when the bottle is opened. Above about 9, the liquid slowly etches aluminum, as colloidal silica does, so keep runs short and rinse promptly.
- Where should used diamond suspension go? Gage poured silica down the sink. PACE's safety data sheet says to follow local regulations (Section 13).

Sources

- Gage's videos: Setting up the vibrator polisher (posted 05-18) and Vibrator polishing + final cleaning + SEM stub mounting (05-21), from digital-alloy-lab#138. His cleaning sheet is in #110.
- Buehler, VibroMet 2 manual MA671635_13.1 (2012), pp. 9-14, ManualsLib.
- G. Vander Voort, "Preparation of Cast Aluminum-Silicon Alloys," Buehler Tech-Notes 3(2): diamond embedding from fine suspensions.
- PACE Technologies: Cleaning, Polycrystalline diamond, and the DIAMAT water-based diamond safety data sheet (revised 07-15-2024).
- Allied High Tech, Polishing Cloths product guide v1.6, p. 1, on removing and re-applying adhesive cloths.
- Edison `LITERATURE_HIGH` searches, 2026-11-03: keep-wet and cleaning and diamond on a vibratory polisher, with full trajectories in the same folder.